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Catalyst bed preheating effects on HTP monopropellant start-up

A. Example, B. Sample - Example Institute of Technology (synthetic fixture)

Abstract

We report start-up tests of a 1 N thruster fed with 90 % hydrogen peroxide. Preheating the silver catalyst bed to 473 K shortened the pressure rise time. The chamber pressure reached 11.8 bar at a preheat temperature of 473 K.

1 Introduction

Cold starts of peroxide thrusters show long ignition delays [1]. Cold starts of peroxide thrusters show long ignition delays [1]. Cold starts of peroxide thrusters show long ignition delays [1]. Cold starts of peroxide thrusters show long ignition delays [1]. Cold starts of peroxide thrusters show long ignition delays [1]. Cold starts of peroxide thrusters show long ignition delays [1].

2 Method

The catalyst bed was heated electrically before each firing [2]. The catalyst bed was heated electrically before each firing [2]. The catalyst bed was heated electrically before each firing [2]. The catalyst bed was heated electrically before each firing [2]. The catalyst bed was heated electrically before each firing [2]. The catalyst bed was heated electrically before each firing [2]. The catalyst bed was heated electrically before each firing [2]. The catalyst bed was heated electrically before each firing [2].

3 Results

The chamber pressure reached 11.8 bar at a preheat temperature of 473 K. Without preheating the rise time was 0.8 s; with preheating it was 0.2 s. The chamber pressure reached 11.8 bar at a preheat temperature of 473 K. Without preheating the rise time was 0.8 s; with preheating it was 0.2 s. The chamber pressure reached 11.8 bar at a preheat temperature of 473 K. Without preheating the rise time was 0.8 s; with preheating it was 0.2 s.

4 Conclusions

Preheating is recommended for repeatable start-up [3]. Preheating is recommended for repeatable start-up [3]. Preheating is recommended for repeatable start-up [3]. Preheating is recommended for repeatable start-up [3]. Preheating is recommended for repeatable start-up [3]. Preheating is recommended for repeatable start-up [3].

References

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